

ALESSANDRO CAROTENUTO

MSc Student, AI & Robotics | Computer and Automation Engineer



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AI / ML, SOFTWARE & ROBOTICS ENGINEER

MSc student in **AI & Robotics** at Sapienza University of Rome. Builds and deploys end-to-end AI/robotics systems across **RL**, **computer vision**, and **multi-robot control**, turning published research into tested, working prototypes.

EDUCATION

Master Degree in Artificial Intelligence & Robotics

2023 - Present

Sapienza, University of Rome

Bachelor Degree in Computer and Automation Engineering

2018 - 2023

Sapienza, University of Rome

High School Diploma for Applied Scientific Studies

2013 - 2018

Lazzaro Spallanzani, Scientific High School

WORK EXPERIENCE

Private Tutor

2017 - Present

Self-Employed, Tivoli, Italy

- Private tutoring in Mathematics, Physics, and Computer Science for high school and university students

SELECTED PROJECTS

Ground2Aerial Synthesis via Transformers

2025

- Built a two-stage **VQGAN-Transformer** pipeline (**PyTorch**, **Transformers (Hugging Face)**, **OpenCV**) for **cross-view image generation** of synthetic aerial images from ground level views, tracked with **Wandb**

Hierarchical RL with World Graph for Navigation

2025

- Built a Hierarchical RL framework (**PyTorch**, **MiniGrid**) with **VAE**-based state discovery and graph-based planning, achieving an **86.8% success rate** on multi-goal navigation tasks

Autonomous Quadrotor Vision-Based Landing

2025 - 2026

- Built a vision-based landing pipeline (**MATLAB/Simulink**, **CoppeliaSim**) with Kalman filtering and **PID control**, achieving **~100% simulated success** and **<0.5 m** tracking error

Collision & Connectivity for Robot Swarms via CBF

2025 - 2026

- Designed, in a team, a distributed safety framework (**MATLAB/Simulink**) using **Control Barrier Functions** and consensus-based connectivity estimation, validated **on a 5-robot swarm across 42 obstacles**

CERTIFICATIONS, VOLUNTEERING, COMPETITIONS

Cisco IT Essentials

Cisco Systems Inc.

2018

Arduino Workshop for Children

Maker Faire Rome

2018

NASA Space Apps Challenge

NASA Space Apps

2019

Sapienza Generative AI Hackathon (2026)

#3 out of 18 teams: developed a CVAE-based music style transfer system, scoring 0.70 harmonic mean.

TECHNICAL SKILLS

Core Programming

Python, C++, MATLAB, TypeScript, JavaScript, Java, Simulink, LaTeX

Libraries & Tools

PyTorch, NumPy, scikit-learn, Matplotlib, SimPy, Git, WandB

Languages

Italian (native), English (B2/C1)

DRIVING LICENCE: Cars: B